

## CLASS ACTIVITY -24/07/2026

### EMPLOYEE SALARY DATASET DATA VISUALIZTIONS

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#### 1.Amounts

# 1. BAR PLOT

```
avg_salary <- employee %>%  
  group_by(Department) %>%  
  summarise(Average_Salary = mean(Monthly_Salary))  
ggplot(avg_salary, aes(x = Department, y = Average_Salary, fill = Department)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Average Monthly Salary by Department",  
    x = "Department",  
    y = "Average Monthly Salary"  
  ) +  
  theme_minimal()
```

```
ggplot(employee,  
  aes(x = Monthly_Salary,  
    y = Department,  
    color = Department)) +  
  geom_dotplot(  
    binaxis = "x",  
    stackdir = "center",  
    dotsize = 0.6  
  ) +  
  labs(
```

```

    title = "Monthly Salary Distribution by Department",
    x = "Monthly Salary",
    y = "Department"
  ) +
  theme_minimal()
# 3. HEATMAP
heatmap_data <- employee %>%
  group_by(Department, Gender) %>%
  summarise(
    Average_Salary = mean(Monthly_Salary),
    .groups = "drop"
  )
ggplot(heatmap_data,
  aes(x = Gender,
    y = Department,
    fill = Average_Salary)) +
  geom_tile() +
  geom_text(aes(label = round(Average_Salary, 0))) +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "Average Salary Heatmap",
    x = "Gender",
    y = "Department",
    fill = "Average Salary"
  ) +

```

theme\_minimal()

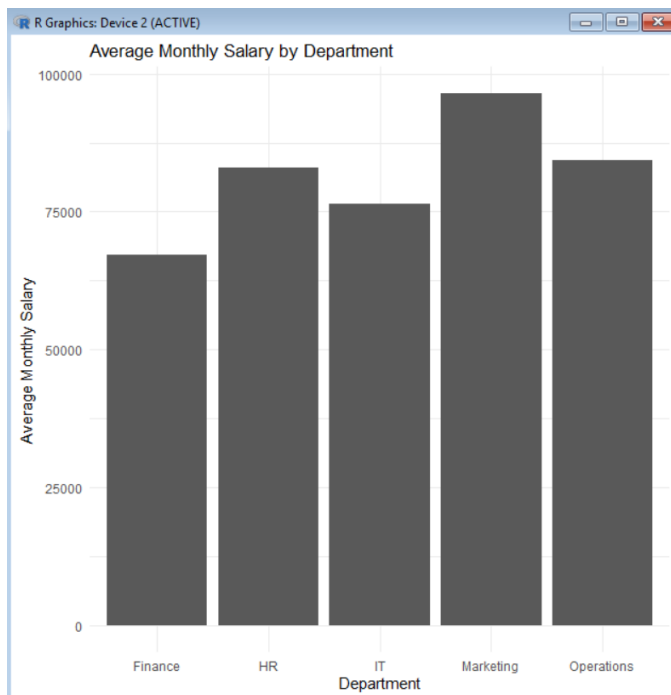
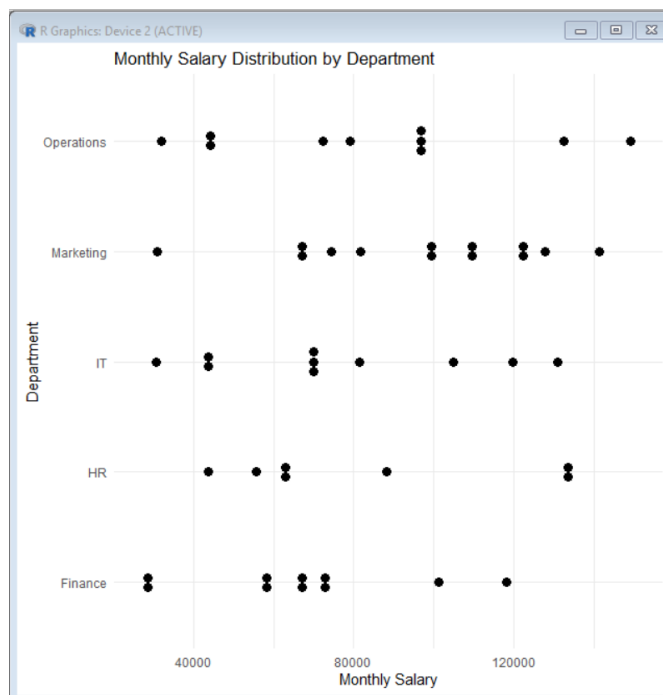


FIG 1. BAR PLOT VISUALIZATION



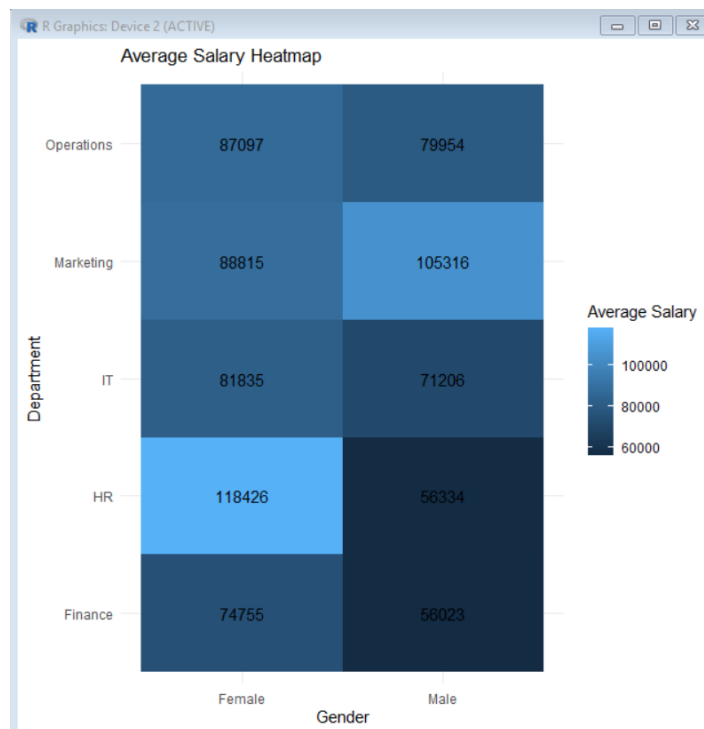


Fig 3.HEATMAP

## VISUALIZING DISTRIBUTIONS :

### # 1. HISTOGRAM

```
ggplot(employee, aes(x = Monthly_Salary)) +
  geom_histogram(
    bins = 10,
    fill = "skyblue",
    color = "darkblue"
  ) +
  labs(
    title = "Distribution of Monthly Salary",
    x = "Monthly Salary",
    y = "Number of Employees"
  ) +
  theme_minimal()
```

## # 2. DENSITY PLOT

```
ggplot(employee,  
  aes(x = Monthly_Salary,  
    fill = Gender,  
    color = Gender)) +  
geom_density(alpha = 0.5) +  
labs(  
  title = "Monthly Salary Density by Gender",  
  x = "Monthly Salary",  
  y = "Density"  
) +  
theme_minimal()
```

## # 3. BOX PLOT

```
ggplot(employee,  
  aes(x = Department,  
    y = Monthly_Salary,  
    fill = Department)) +  
geom_boxplot() +  
labs(  
  title = "Monthly Salary Distribution by Department",  
  x = "Department",  
  y = "Monthly Salary"  
) +  
theme_minimal()
```

## # 4. RIDGELINE PLOT

```
ggplot(employee,  
  aes(x = Monthly_Salary,  
    y = Department,
```

```

    fill = Department)) +
  geom_density_ridges(alpha = 0.7) +
  labs(
    title = "Salary Distribution by Department",
    x = "Monthly Salary",
    y = "Department"
  ) +
  theme_ridges()

```

```

# =====
# 5. VIOLIN PLOT
# =====

```

```

ggplot(employee,
  aes(x = Education_Level,
    y = Monthly_Salary,
    fill = Education_Level)) +
  geom_violin() +
  geom_boxplot(
    width = 0.1,
    fill = "white"
  ) +
  labs(
    title = "Salary Distribution by Education Level",
    x = "Education Level",
    y = "Monthly Salary"
  ) +

```

theme\_minimal()

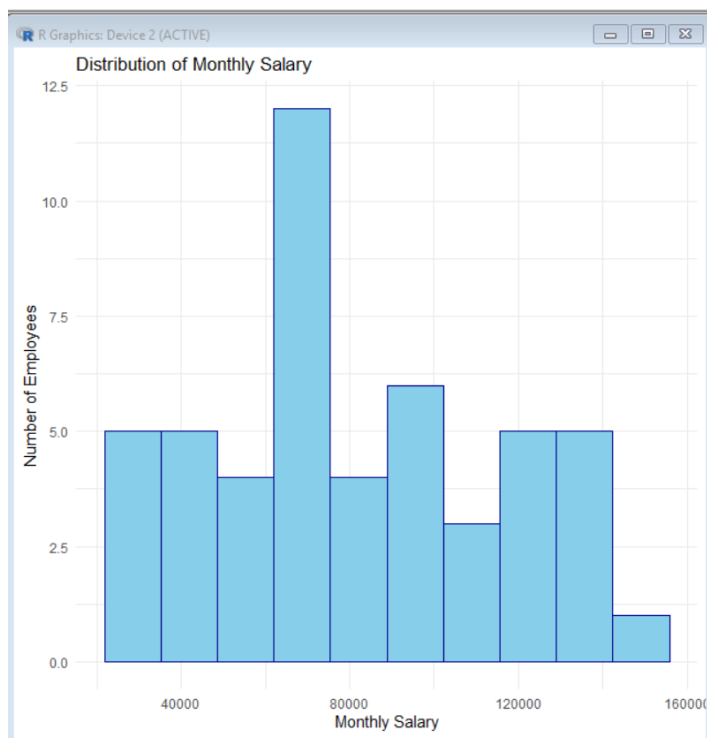


Fig 4. Histogram

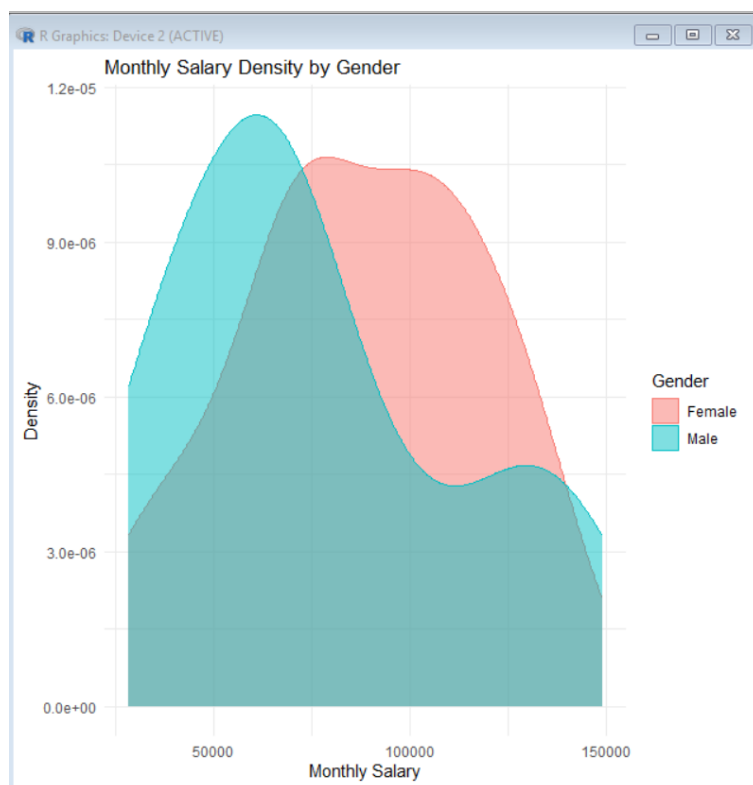


Fig 5. Density plot

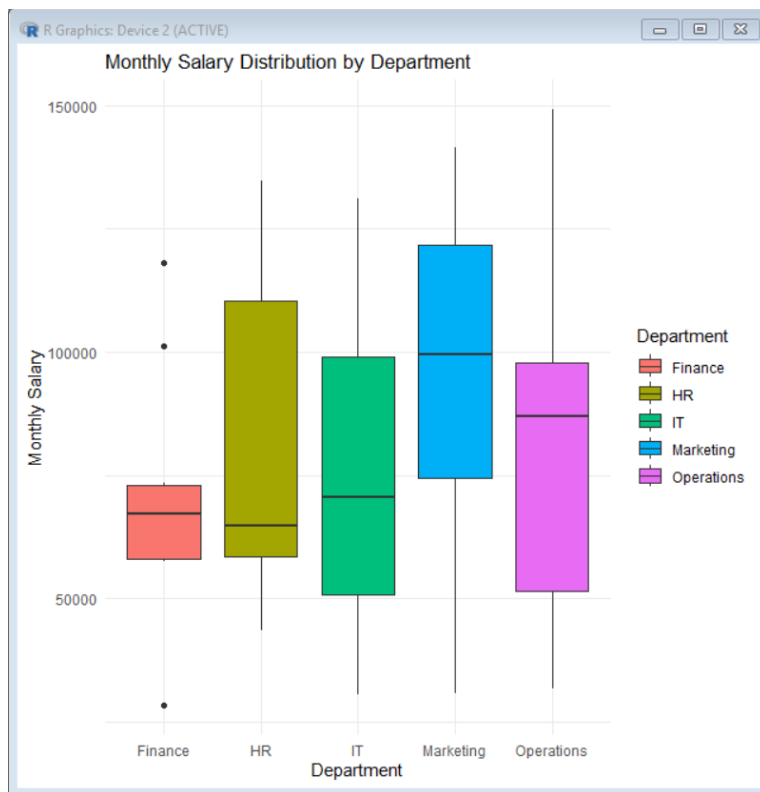


Fig 6. Box plot

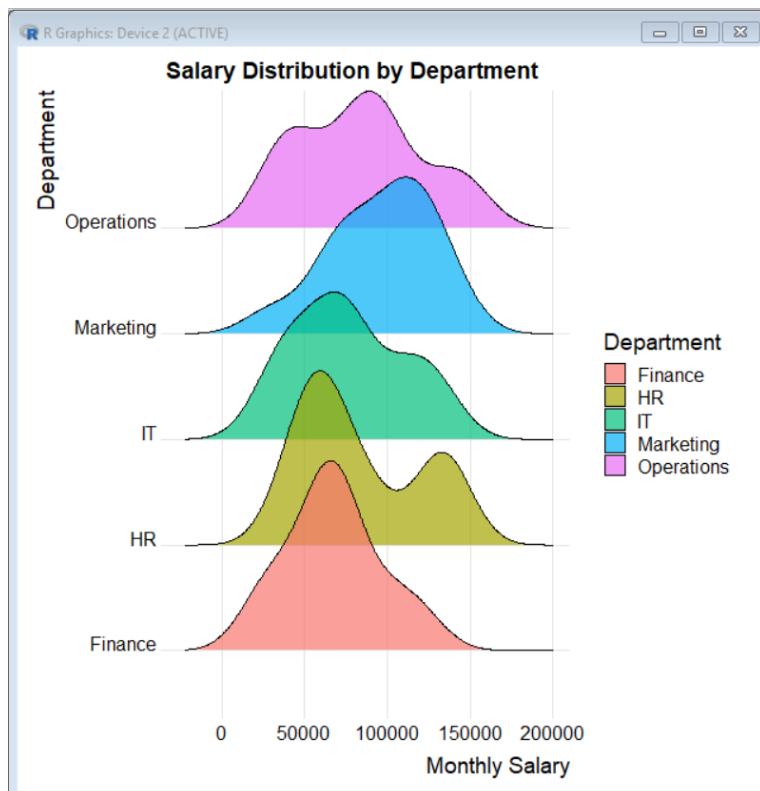


Fig 7. Ridgeline plot



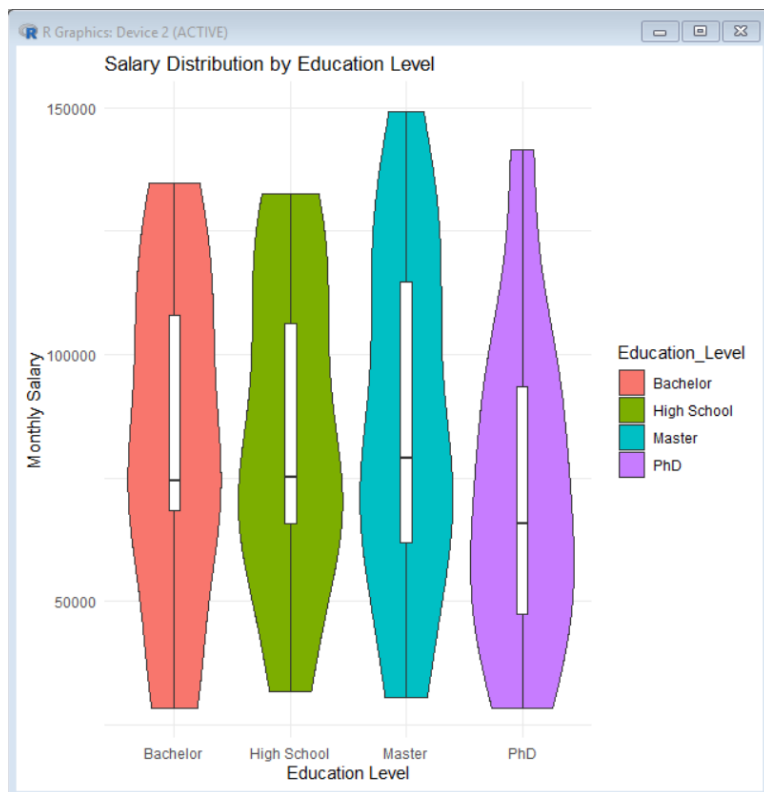


Fig 8.Violin plot

## Visualizing Proportions

# =====

### # 1. PIE CHART

```
department_count <- employee %>%
```

```
  count(Department)
```

```
ggplot(department_count,
```

```
  aes(x = "",
```

```
    y = n,
```

```
    fill = Department)) +
```

```
geom_bar(
```

```
  stat = "identity",
```

```
  width = 1,
```

```
  color = "white"
```

```
) +
```

```
coord_polar("y") +
```

```

labs(
  title = "Percentage of Employees by Department"
) +
theme_void()

# =====
# 2. STACKED BAR CHART
# =====

```

```

ggplot(employee,
  aes(x = Department,
    fill = Gender)) +
geom_bar() +
labs(
  title = "Gender Distribution by Department",
  x = "Department",
  y = "Number of Employees"
) +
theme_minimal()

# =====
# 3. TREEMAP
# =====

```

```

treemap_data <- employee %>%
  group_by(Department) %>%
  summarise(
    Total_Salary = sum(Monthly_Salary)
  )

```

```

ggplot(treemap_data,
  aes(
    area = Total_Salary,
    fill = Department,
    label = paste(
      Department,
      "\nSalary:",
      Total_Salary
    )
  )) +
  geom_treemap() +
  geom_treemap_text(
    color = "white",
    place = "centre"
  ) +
  labs(
    title = "Total Salary by Department"
  ) +
  theme_minimal()

```

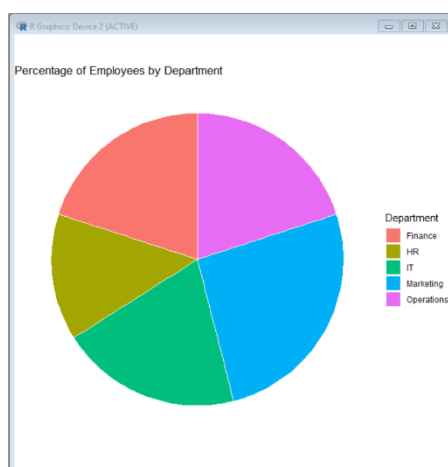


Fig 9. Pie chart

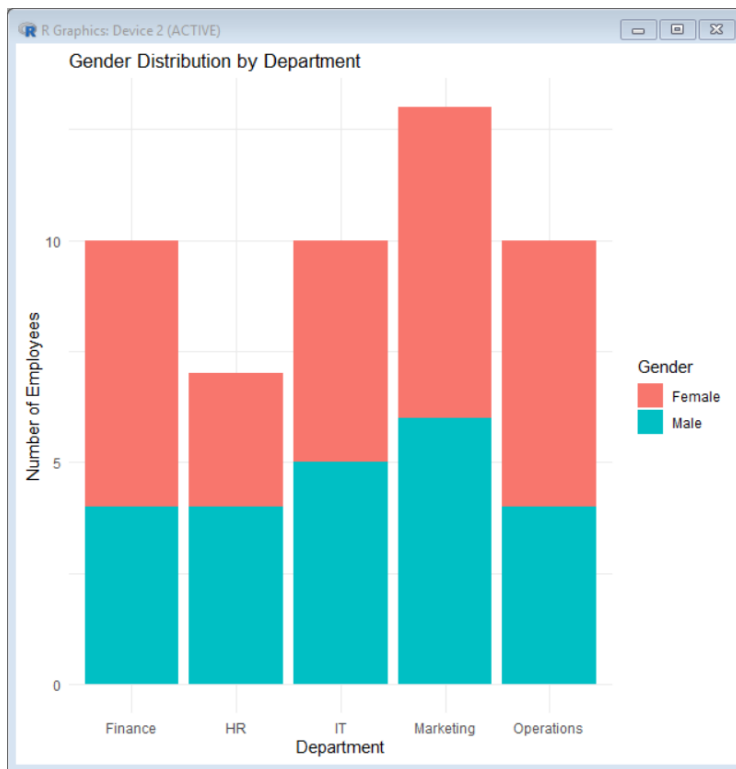


Fig 10.Stacked Bar chart

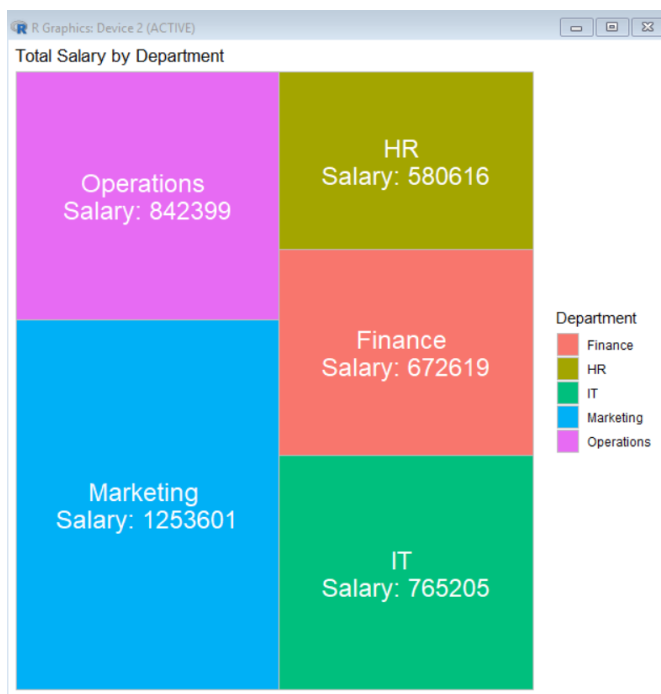


Fig 11.Treemap

## X-Y Relationships

```
# =====
```

```
# 1. SCATTER PLOT
```

```
# =====
```

```
ggplot(employee,  
  aes(  
    x = Experience_Years,  
    y = Monthly_Salary,  
    color = Department  
  )) +  
  geom_point(size = 3) +  
  geom_smooth(  
    method = "lm",  
    se = FALSE,  
    color = "black"  
  ) +  
  labs(  
    title = "Experience vs Monthly Salary",  
    x = "Experience in Years",  
    y = "Monthly Salary",  
    color = "Department"  
  ) +  
  theme_minimal()
```

```
# =====
```

```
# 2. BUBBLE CHART
```

```
# =====
```

```

ggplot(employee,
  aes(
    x = Experience_Years,
    y = Monthly_Salary,
    size = Age,
    fill = Department
  )) +
geom_point(
  alpha = 0.7,
  shape = 21,
  color = "black"
) +
labs(
  title = "Bubble Chart: Experience, Salary and Age",
  x = "Experience in Years",
  y = "Monthly Salary",
  size = "Age",
  fill = "Department"
) +
theme_minimal()

```

### # 3. HEXBIN PLOT

```

# =====

ggplot(employee,
  aes(
    x = Age,
    y = Monthly_Salary
  )) +
geom_hex() +

```

```

scale_fill_gradient(
  low = "lightgreen",
  high = "darkgreen"
) +
labs(
  title = "Hexbin Plot: Age vs Monthly Salary",
  x = "Age",
  y = "Monthly Salary",
  fill = "Count"
) +
theme_minimal()

```

```
# =====
```

```
# 4. 2D DENSITY MAP
```

```
# =====
```

```

ggplot(employee,
  aes(
    x = Experience_Years,
    y = Monthly_Salary
  )) +
geom_density_2d_filled() +
labs(
  title = "2D Density Map: Experience vs Monthly Salary",
  x = "Experience in Years",
  y = "Monthly Salary"
) +

```

theme\_minimal()

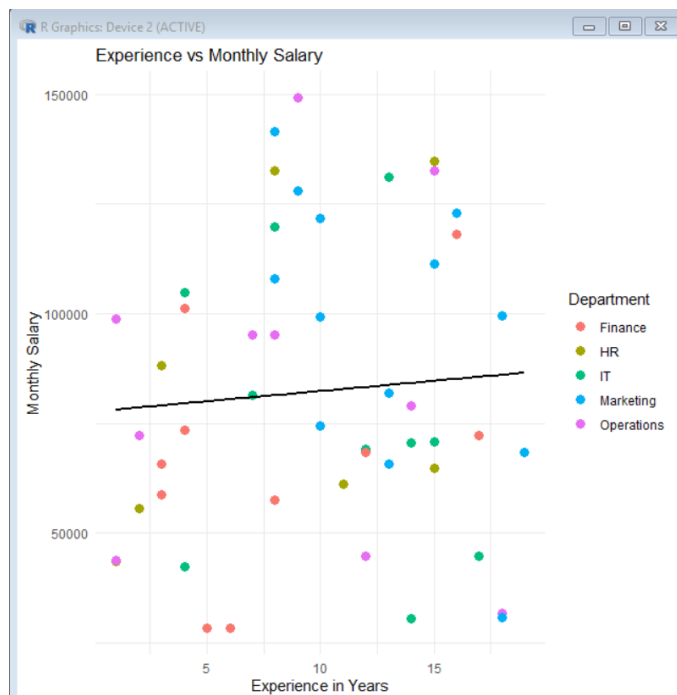


Fig 12.Scatter plot



Fig 13.Bubble chart



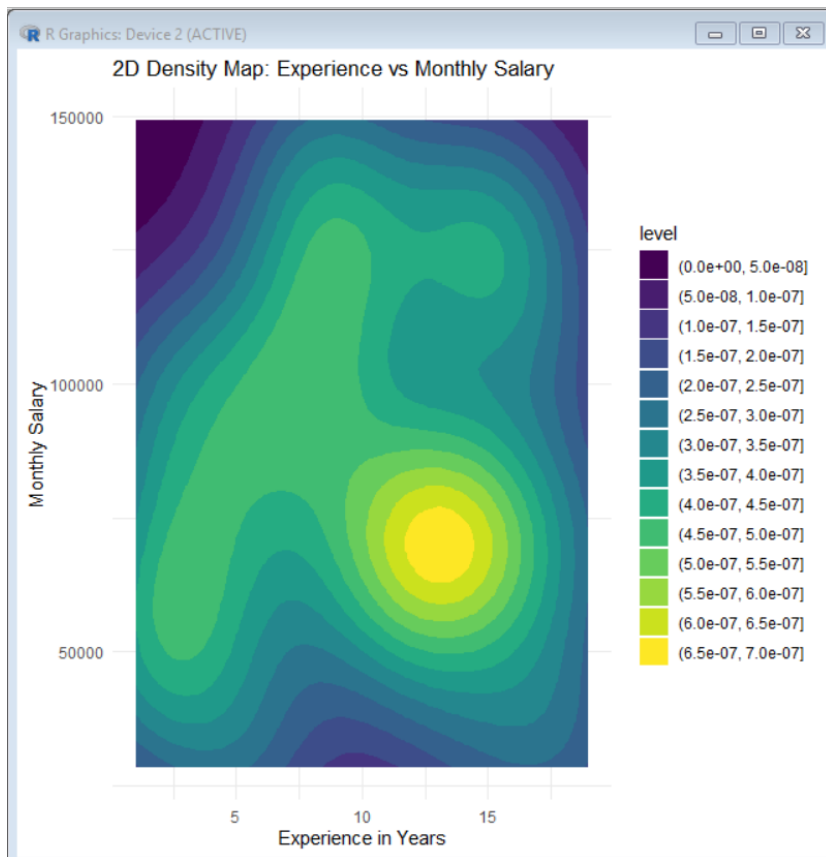


Fig 14. 2D Density Map